SHUBHAM KUMAR

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SUMMARY

B.Tech CSE student (Class of 2027) focused on developer tooling and CI/CD. Projects include a pull-request security checker and GitHub App in Python, a data pipeline with schema validation and an LLM fallback, and a MERN community platform. Strong in DSA, OOP, OS, networking and DBMS; looking for an internship on a platform or developer-experience team.

EDUCATION

Lovely Professional University, Phagwara, Punjab Aug 2023 – Present
B.Tech, Computer Science and Engineering | CGPA 7.95 | Expected graduation: 2027

Trident Public School, Muzaffarpur, Bihar | Intermediate (Class XII) Apr 2021 – Mar 2022

SKILLS

Languages: Python, Java, C++, JavaScript, C, PHP
CS Fundamentals: DSA, Object-Oriented Design, Operating Systems, Computer Networks, DBMS, Complexity Analysis
Backend & Web: FastAPI, Node.js, React, Jinja2, httpx, HTML/CSS, Bootstrap
CI/CD & Platform: GitHub Actions, Docker, OPA (policy as code), Copier templates, pytest, ruff, Git
Security & Supply Chain: gitleaks, Trivy, Syft (CycloneDX SBOM), HMAC webhook verification, RBAC
Data & LLM: Pydantic, SQLite, MySQL, MongoDB, LLM APIs (Groq), structured-output validation

PROJECTS

ShipGuard: Pull-Request Security & Supply-Chain Automation | GitHub Sep 2026 – Present
Python 3.12 · FastAPI · GitHub Actions · OPA · Docker · Copier

- Developed a Python CLI that runs three scanners (gitleaks, Trivy, Syft) on PRs and fails at the fail_on level in .shipguard.yml. Policies are in OPA, and a reusable GitHub Actions workflow runs lint, tests, policy and SBOM, reporting a Check Run.
- Built the FastAPI webhook for the GitHub App. It verifies the HMAC signature before parsing anything, loads policy from the base SHA rather than the PR, and checks out the PR code without running it. The checker runs with credentials removed from its environment to limit what a malicious PR can reach.
- Integrated an LLM reviewer that validates model output against a schema and comments only on confident findings on added diff lines, scored on 30 synthetic evaluation cases (10 clean) with shipguard eval. Also made a Copier template for FastAPI services with AGENTS.md, CI, tests and a digest-pinned, non-root Dockerfile.

Job Ingestion Pipeline | GitHub | Live Jul 2026 – Aug 2026
Python · httpx · tenacity · Pydantic · SQLite · FastAPI · Jinja2 · Groq

- Implemented a Python pipeline that pulls from a live public API using httpx and tenacity, with user-agent rotation, random delays and an exponential-backoff circuit breaker. Upserts are idempotent, so reruns don't duplicate rows.
- Validated every record against strict Pydantic schemas, with an optional LLM fallback parser (Groq) that repairs malformed ones and logs latency and token usage on each call so the cost stays visible.
- Added a rule-based enrichment step (30+ regex skill extractors, seniority level, remote/hybrid/onsite) and a FastAPI + Jinja2 dashboard showing pipeline health and the log of LLM repairs.

EKYAM | GitHub | Live Jan 2026 – Mar 2026
MongoDB · Express · React · Node.js

- Created a community platform with member-led moderation, per-community configuration and automated admin workflows.
- Wrote the events module, with RSVP tracking and automatic participation logs.
- Secured the resource-sharing section with role-based access control (RBAC), which protects document uploads and establishes fine-grained permissions across multi-level communities.

TRAINING & CERTIFICATIONS

- **CSE Pathshala** (Jun 2025 – Jul 2025): 32-day C++ program on OOP and DSA; built 5+ projects with trees, graphs and hash tables.
- **IBM, Hardware and Operating Systems (with Honours)** (Aug 2024 – Sep 2024): OS configuration and troubleshooting, hardware including GPUs, secure IoT integration and VR/AR security.
- **Oracle Cloud Infrastructure 2025 Certified AI Foundations Associate**, May 2026 | Credential
- Computational Theory: Language Principle & Finite Automata Theory (Infosys), Aug 2025 | Certificate
- Social Networks (NPTEL, Government of India), Apr 2025 | Certificate
- The Bits and Bytes of Computer Networking (Google), Sep 2024 | Certificate

ACHIEVEMENTS

- Solved 500+ algorithmic problems in C++ on Leet Code with 1750+ contest rating
- 2nd Runner-up, CYPHERTHON coding challenge (Mar 2025)
- Winner, CPEC Club Debate Competition (Nov 2024)
- Ran CPEC Club's social media; ranked #1 among all CPEC clubs for three consecutive months (Sep 2023 – Mar 2024)